Coordinated Pause Escrow: POC document

Goal

- Prove the cryptographic core works: whoever locked a commitment cannot change it, and nobody can peek before everyone has locked.
- Two competing companies each lock a commitment to pause development. The commitments are revealed together, and only after both are locked.

Components

- Commitment Generator (each side): creates a locked commitment and sends only its hash.
- Escrow Coordinator: stores commitments only, never their content.
- Reveal Engine: reveals only when both sides have locked and been verified.

How it works

- Pedersen commitment: C = g^m · h^r modulo a 2048-bit prime. The value r is random, so the commitment reveals nothing about the message (hiding).
- Opening the commitment to a different message would require solving a discrete logarithm, so nothing can be changed after locking (binding).
- Escrow states: COLLECTING, ALL_LOCKED, TRIGGERED or ABORTED.

What is tested (four scenarios)

- Honest run: both sides lock and open, and the simulated pause event fires.
- Peek attempt: a reveal request before everyone has locked is refused.
- Cheat attempt: opening to different content, or locking twice, is rejected.
- Vanished party: after a timeout the escrow aborts and nothing is revealed (liveness).

POC boundaries

- In scope: the cryptographic protocol core, two simulated nodes and a simulated pause event.
- Out of scope: a real organizational trust layer, who operates the escrow, and real communication between companies.
- The code uses only the Python standard library and also runs in the browser.

Ethical question

- A protocol meant to let competitors coordinate without breaching antitrust law could itself become a tool for cartel coordination. What prevents misuse of the protocol itself? This must be answered in the specification.

Conclusion and hand-off to specification

- The cryptographic core is proven.
- Liveness: the run shows an abort after a timeout. The specification will define the timeout and its consequences.
- Who operates the escrow in the real world: a trusted, agreed authority is required. Left open for the specification.